Derivation of Einstein Field Equation By Hilbert Action

first of all we know that $I = \int_M f \sqrt{|g|}\, d^4x$

I = action in physics (scalar)

M = represent 4 dim space time manifold

f = density of Lagrange

↳ For instance; $f = \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi)$

$$f = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

$$f = \frac{1}{16\pi G} R \qquad R: \text{Ricci scalar}$$

$\sqrt{|g|}$ = integration measure

d^4x = differential Volume element $d^4x = dx^0 dx^1 dx^2 dx^3$

$$d^4V = \sqrt{|g|}\, d^4x$$

in Minkow action = $I = \int f\, d^4x \Rightarrow d^4x = \sqrt{|g|}\, d^4x$

$$I = \int_M f \sqrt{|g|}\, d^4x$$

$\Rightarrow$ we can take $f = \frac{1}{16\pi G} R$

There fore $I_{EH} = \frac{1}{16\pi G} \int_M R \sqrt{|g|}\, d^4x$ is called

Einstein Hilbert action

now we vary the action to reach equation of motion

$$\delta I_{EH} = \frac{1}{16\pi G} \delta \int_M R \sqrt{|g|}\, d^4x$$

$$\Rightarrow \frac{1}{16\pi G} \left\{ \underbrace{\int_M \delta R \sqrt{|g|}\, d^4x}_{(i)} + \underbrace{\int_M R\, \delta\sqrt{|g|}\, d^4x}_{(ii)} \right.$$

$$(ii) \quad \int_M R\, \delta |g|^{1/2}\, d^4x \Rightarrow \delta\sqrt{|g|} = ?$$

$$\Rightarrow \quad \det(g_{\mu\nu}) = g \quad \text{and} \quad \sqrt{|g|}$$

we know that Liner Algebra identity

$$\delta(\det A) = (\det A)\, Tr(A^{-1}\, \delta A) \qquad A = g_{\mu\nu} \quad \text{Thus}$$

$$\delta g = g\, g^{\mu\nu} \delta g_{\mu\nu}$$

$$\delta\sqrt{|g|} = \frac{1}{2\sqrt{|g|}} \delta |g|$$

$$\delta |g| = |g|\, g^{\mu\nu} \delta g_{\mu\nu}$$

$$\delta\sqrt{|g|} = \frac{1}{2\sqrt{|g|}} \left(|g|\, g^{\mu\nu} \delta g_{\mu\nu} \right)$$

$$\Rightarrow \delta\sqrt{|g|} = \frac{1}{2} \left(\sqrt{|g|}\, g^{\mu\nu} \delta g_{\mu\nu} \right)$$

(i) $\int_M \delta R \sqrt{|g|}\, d^4x \qquad R =: g^{\mu\nu} R_{\mu\nu}$

(i)

$$\delta R = \delta\left(g^{\mu\nu} R_{\mu\nu}\right) = \underset{(a)}{\delta g^{\mu\nu} R_{\mu\nu}} + \underset{(b)}{g^{\mu\nu}\delta R_{\mu\nu}}$$

(a) $\delta g^{\mu\nu} R_{\mu\nu}$ is directly

(b) $g^{\mu\nu}\delta R_{\mu\nu}$ is important!

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu} \Rightarrow R^{\lambda}{}_{\mu\lambda\nu} = \partial_\lambda \Gamma^{\lambda}_{\mu\nu} - \partial_\mu \Gamma^{\lambda}_{\mu\lambda} + \Gamma^{\lambda}_{\sigma\lambda}\Gamma^{\sigma}_{\mu\nu} - \Gamma^{\lambda}_{\sigma\nu}\Gamma^{\sigma}_{\mu\lambda}$$

only christoffel symbols are varied because ∂ operator is linear.

$$(*)\ \delta R_{\mu\nu} = \partial_\lambda \delta\Gamma^{\lambda}_{\mu\nu} - \partial_\mu \delta\Gamma^{\lambda}_{\mu\lambda} + (\delta\Gamma^{\lambda}_{\sigma\lambda})\Gamma^{\sigma}_{\mu\nu} + \Gamma^{\lambda}_{\sigma\lambda}(\delta\Gamma^{\sigma}_{\mu\nu}) - (\delta\Gamma^{\lambda}_{\sigma\nu})\Gamma^{\sigma}_{\mu\lambda} - \Gamma^{\lambda}_{\sigma\nu}(\delta\Gamma^{\sigma}_{\mu\lambda})$$

Definition of Covariant Derivative;

$$\nabla_\alpha T^{\lambda}_{\mu\nu} = \partial_\alpha T^{\lambda}_{\mu\nu} + \Gamma^{\lambda}_{\sigma\alpha} T^{\alpha}_{\mu\nu} - \Gamma^{\sigma}_{\mu\alpha} T^{\lambda}_{\sigma\nu} - \Gamma^{\sigma}_{\nu\alpha} T^{\lambda}_{\mu\sigma}$$

$$T^{\lambda}_{\mu\nu} =: \delta\Gamma^{\lambda}_{\mu\nu}$$

$$\Rightarrow \nabla_\lambda(\delta\Gamma^{\lambda}_{\mu\nu}) = \partial_\lambda(\delta\Gamma^{\lambda}_{\mu\nu}) + \Gamma^{\lambda}_{\sigma\lambda}\delta\Gamma^{\sigma}_{\mu\nu} - \Gamma^{\sigma}_{\mu\lambda}\delta\Gamma^{\lambda}_{\sigma\nu} - \Gamma^{\sigma}_{\nu\lambda}\delta\Gamma^{\lambda}_{\mu\sigma} \text{ --- } (1)$$

$$\nabla_\nu(\delta\Gamma^{\lambda}_{\mu\lambda}) = \partial_\nu(\delta\Gamma^{\lambda}_{\mu\lambda}) + \Gamma^{\lambda}_{\sigma\nu}\delta\Gamma^{\sigma}_{\mu\lambda} - \Gamma^{\sigma}_{\mu\nu}\delta\Gamma^{\lambda}_{\sigma\lambda} - \Gamma^{\sigma}_{\lambda\nu}\delta\Gamma^{\lambda}_{\mu\sigma} \text{ --- } (2)$$

(1)−(2)

$$= \partial_\lambda(\delta\Gamma^\lambda_{\mu\nu}) - \partial_\nu(\delta\Gamma^\lambda_{\mu\lambda}) =$$

$$\partial_\lambda(\delta\Gamma^\lambda_{\mu\nu}) - \partial_\nu(\delta\Gamma^\lambda_{\mu\lambda}) +$$

$$+\left[\Gamma^\lambda_{\sigma\lambda}\delta\Gamma^\sigma_{\mu\nu} - \Gamma^\sigma_{\mu\lambda}\delta\Gamma^\lambda_{\sigma\nu} - \Gamma^\sigma_{\nu\lambda}\delta\Gamma^\lambda_{\mu\sigma}\right] -$$

$$-\left[\Gamma^\lambda_{\sigma\nu}\delta\Gamma^\sigma_{\mu\lambda} - \Gamma^\sigma_{\mu\nu}\delta\Gamma^\lambda_{\sigma\lambda} - \Gamma^\sigma_{\lambda\nu}\delta\Gamma^\lambda_{\mu\sigma}\right]$$

conclusion is (★) equation.

The Result;

$$\delta R_{\mu\nu} = \nabla_\lambda(\delta\Gamma^\lambda_{\mu\nu}) - \nabla_\nu(\delta\Gamma^\lambda_{\mu\lambda})$$

$$\delta R = R_{\mu\nu}\delta g^{\mu\nu} + g^{\mu\nu}\delta R_{\mu\nu}$$

$$\delta\Gamma^\lambda_{\mu\nu} = \frac{1}{2}g^{\lambda\sigma}(\partial_\mu\delta g_{\nu\sigma} + \partial_\nu\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\nu})$$

$$\delta\Gamma^\lambda_{\mu\lambda} = \frac{1}{2}g^{\lambda\sigma}(\partial_\mu\delta g_{\lambda\sigma} + \partial_\lambda\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\lambda})$$

$$= g^{\mu\nu}\delta R_{\mu\nu} = \nabla_\alpha V^\alpha$$

$$\delta R = R_{\mu\nu}\delta g^{\mu\nu} + \nabla_\alpha V^\alpha$$

If we are in approximity of manifold's border

$$\delta g_{\mu\nu} = 0 \rightarrow \nabla_\alpha V^\alpha = 0$$

$$\delta\sqrt{|g|} = \frac{1}{2}\left(\sqrt{|g|}\, g^{\mu\nu}\,\delta g_{\mu\nu}\right)$$

$$\delta R = R_{\mu\nu}\,\delta g^{\mu\nu} + \cancelto{0}{\nabla_\alpha V^\alpha}$$

$$\int \delta R \sqrt{|g|}\, d^4x + \int R\,\delta\sqrt{|g|}\, d^4x$$

$$\Rightarrow \int R_{\mu\nu}\,\delta g^{\mu\nu}\sqrt{|g|}\, d^4x + \frac{1}{2}\int R\sqrt{|g|}\, g^{\mu\nu}\,\delta g_{\mu\nu}$$

$$\Rightarrow \delta\sqrt{|g|} = +\frac{1}{2}\sqrt{|g|}\, g^{\mu\nu}\,\delta g_{\mu\nu} = -\frac{1}{2}\sqrt{|g|}\, g_{\mu\nu}\,\delta g^{\mu\nu} \quad \star$$

$$\Rightarrow \int R_{\mu\nu}\,\delta g^{\mu\nu}\sqrt{|g|}\, d^4x - \frac{1}{2}\int R\sqrt{|g|}\, g_{\mu\nu}\,\delta g^{\mu\nu}\, d^4x$$

$$\Rightarrow \int \delta g^{\mu\nu}\left(R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu}\right)\sqrt{|g|}\, d^4x$$

$$\Rightarrow \delta I_{EH} = \frac{1}{16\pi G}\int \left[R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu}\right]\delta g^{\mu\nu}\sqrt{|g|}\, d^4x$$

if we include matter,

$$\delta I_{mat} = -\frac{1}{2}\int T_{\mu\nu}\,\delta g^{\mu\nu}\sqrt{|g|}\, d^4x \Rightarrow T_{\mu\nu} = \frac{-2}{\sqrt{|g|}}\frac{\delta I_{mat}}{\delta g^{\mu\nu}}$$

$$\delta I_{EH} + \delta I_{mat} = 0$$

$I_{mat} = \int \mathcal{L}_{mat} \cdot \sqrt{|g|}\, dx^4$ $\mathcal{L}_{mat}$ is Lagrange density of matter (i.e. EM field $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$)

$\delta I_{mat} + \delta I_{EH} = 0$ (stationary action)

$\delta I_{mat} = \delta \int \mathcal{L}_{mat} \cdot \sqrt{|g|}\, dx^4 \Rightarrow$

$$\delta I_{mat} = \int \underbrace{\delta \mathcal{L}_{mat} \sqrt{|g|}}\, dx^4 + \int \mathcal{L}_{mat} \overbrace{\delta \sqrt{|g|}}\, dx^4$$

$$\delta \mathcal{L}_{mat} = \frac{\partial \mathcal{L}_{mat}}{\partial g^{\mu\nu}} \delta g^{\mu\nu} \qquad -\frac{1}{2}\sqrt{|g|}\, g_{\mu\nu}\, \delta g^{\mu\nu}$$

$$\Rightarrow \frac{\partial \mathcal{L}_{mat}}{\partial g^{\mu\nu}} \delta g^{\mu\nu} - \frac{1}{2}\sqrt{|g|}\, g_{\mu\nu}\, \delta g^{\mu\nu}$$

$$\int \left[\frac{\partial \mathcal{L}_{mat}}{\partial g^{\mu\nu}} - \frac{1}{2}\sqrt{|g|}\, g_{\mu\nu}\right] \delta g^{\mu\nu}\, dx^4$$

$$T_{\mu\nu} = \frac{-2}{\sqrt{|g|}} \frac{\delta(\sqrt{|g|}\,\mathcal{L}_{mat})}{\delta g^{\mu\nu}}$$

$$\underset{(i)}{\frac{-2}{\sqrt{|g|}} \frac{\delta(\sqrt{|g|}\,\mathcal{L}_{mat})}{\delta g^{\mu\nu}}} \quad \text{is} \quad \underset{(ii)}{T_{\mu\nu} = \frac{-2}{|\sqrt{g}|}\left[\frac{\partial \mathcal{L}_{mat}}{\partial g^{\mu\nu}} - \frac{1}{2}\mathcal{L}_{mat}\, g_{\mu\nu}\right]}$$

(i) equals to (ii) if;

(i) takes specific coefficent

$$\delta I_{EH} = \frac{1}{16\pi G} \int \left[R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right] \delta g^{\mu\nu} \sqrt{|g|}\, d^4x$$

$$\delta I_{mat} = -\frac{1}{2} \int T_{\mu\nu}\, \delta g^{\mu\nu} \sqrt{|g|}\, d^4x$$

$$\delta I_{EH} + \delta I_{mat} = \left\{ \int d^4x\, \delta g^{\mu\nu} \sqrt{|g|} \left(\underbrace{\frac{1}{16\pi G} \left\{ R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right\} - \frac{1}{2} T_{\mu\nu}}_{=0} \right) \right\} = 0$$

$$\Rightarrow \frac{1}{2} T_{\mu\nu} = \frac{1}{16\pi G} \left[R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right]$$

$$\Rightarrow \frac{16\pi G}{2} T_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$$

<u>Note</u> $\frac{1}{16\pi G}$ is influenced by experimental observation